CSE 1141 - COMPUTER PROGRAMMING I
Programming Assignment # 1
DUE DATE: 21/10/2019 - 23:00 (No extension)

1. In this program, you are given the x and y coordinates of two points as (x_1, y_1) and (x_2, y_2). Assume that a line passes through these points, and you are asked to give the equation of the line as the output of your program. You should also find the slope of the line.

- The slope will be calculated as: $slope = \frac{(y2-y1)}{(x2-x1)}$
- The equation of a line is typically written as y=mx+b where m is the slope and b is the y-intercept.
- To find b, use any (x,y) pair, put the values in the equation and solve for b.
- Test your program to be sure it works with different pairs of points.

Example Run

```
Enter first point (x1,y1): 3 5
Enter second point (x2,y2):-1 3
The slope of the equation is 0.5
The equation is y=(0.5)x+(3.5)
```

Important Note: You should use proper data types for your variables.

2. Suppose you are asked to develop a program to help cashiers in a market store. The program will take billing amount and customers payment that is the money given to the cashier as input. The program is required to show the cashier how to pay change with a report listing the monetary equivalent in the maximum number of 100TLs, 50TLs, 20Tls, 10TLs, 5TLs, 1TLs, and 50Krs, 25Krs, 10Krs, 5Krs, 1Krs, in the given order.

Example Run

```
Enter the billing amount: 103.15
Enter the money given: 500
1 200 TL
1 100 TL
1  50 TL
2  20 TL
0  10 TL
1   5 TL
1   1 TL
1  50 KR
1  25 KR
1  10 KR
0   5 KR
0   1 KR
```

Important Note: The output of your program must be the same as the example above. Your inputs should be defined as double values.

Submission Instructions:

Please zip and submit all your files using filename YourNumberHW1.zip to Canvas system (under Assignments tab). Your zip file should contain the following:

1. Java source code for Problem 1 (Pro1_yournumber.java)
2. Java class file for Problem 1 (Pro1_ yournumber.class)
3. Java source code for Problem 2 (Pro2_ yournumber.java)
4. Java class file for Problem 2 (Pro2_ yournumber.class)

Notes:

1. Write a comment at the beginning of each program to explain the purpose of the program. Write your name and student ID as a comment. Include necessary comments to explain your actions.
2. Select meaningful names for your variables.
3. You are allowed to use the materials that you have learned in lectures & labs.
4. Do not use the things that you did not learn in the course.
5. Each student should submit his/her own homework. You can discuss with your peers about the homework but you are not allowed to exchange code or pseudocode. This also applies to material found on the web. Should some submitted homework assignments be identical or suspected to be identical, all involved parties will get a grade of **ZERO** from all homework. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties.
6. No late submission will be accepted.

Grading:

Problem 1 (32 points)

- Reading four input values from the user (10 points)
- Finding the slope of the line (10 points)
- Printing the output in $y = mx + b$ format (12 points)

Problem 2 (58 points)

- Reading two input values from the user (10 points)
- Printing the amount of each cash (12*4=48 points)

Submission format (10 points)

- 2 java files + 2 class files
- Make sure that your class files can be executed on another computer.
- Make sure that the input/output of your program must be the same with the examples above (all informative strings & spaces).
- Comments are necessary!